

EEET202

THINES & TRANSFORMERS

MODULE:2

ecture:6 – Voltage Build-Up

e performance characteristics of DC generators

Prepared By,

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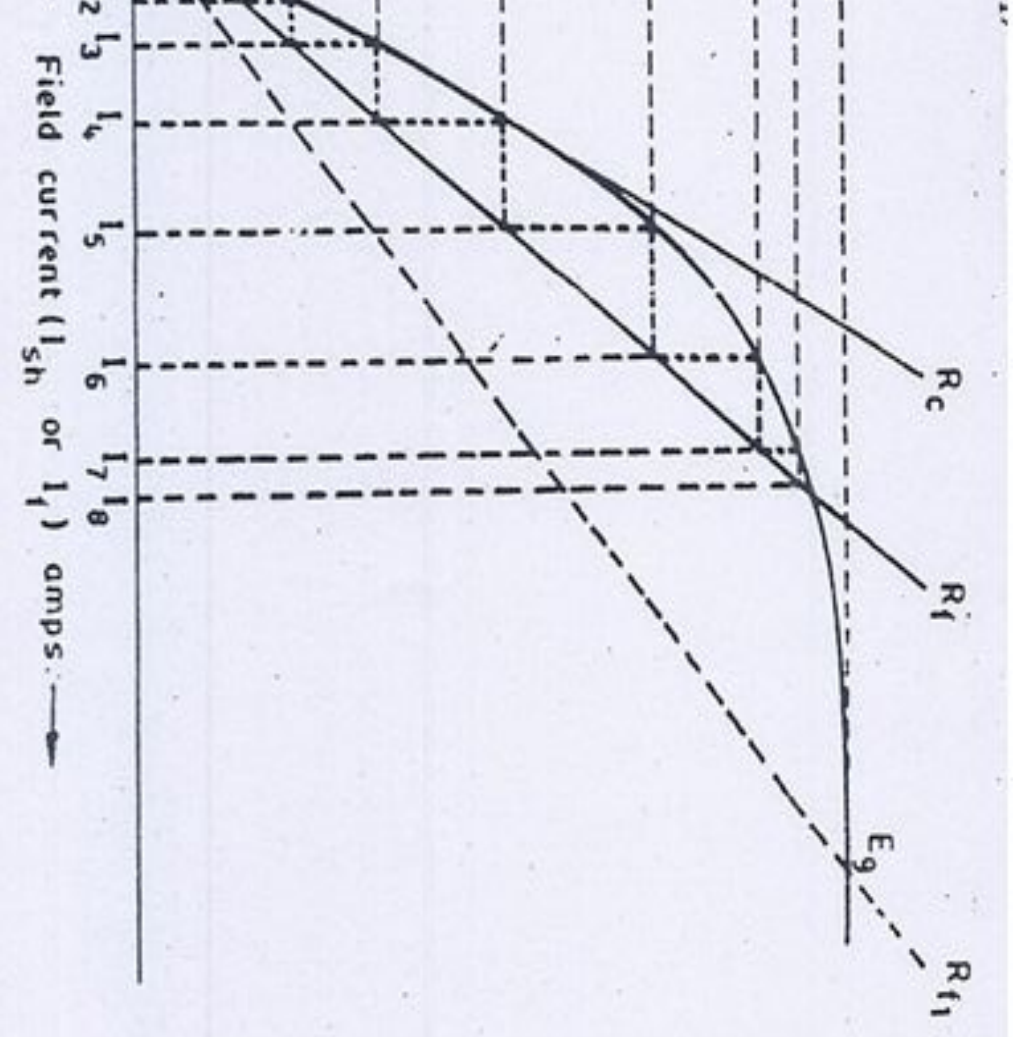
Asst. Professor

EEE Department

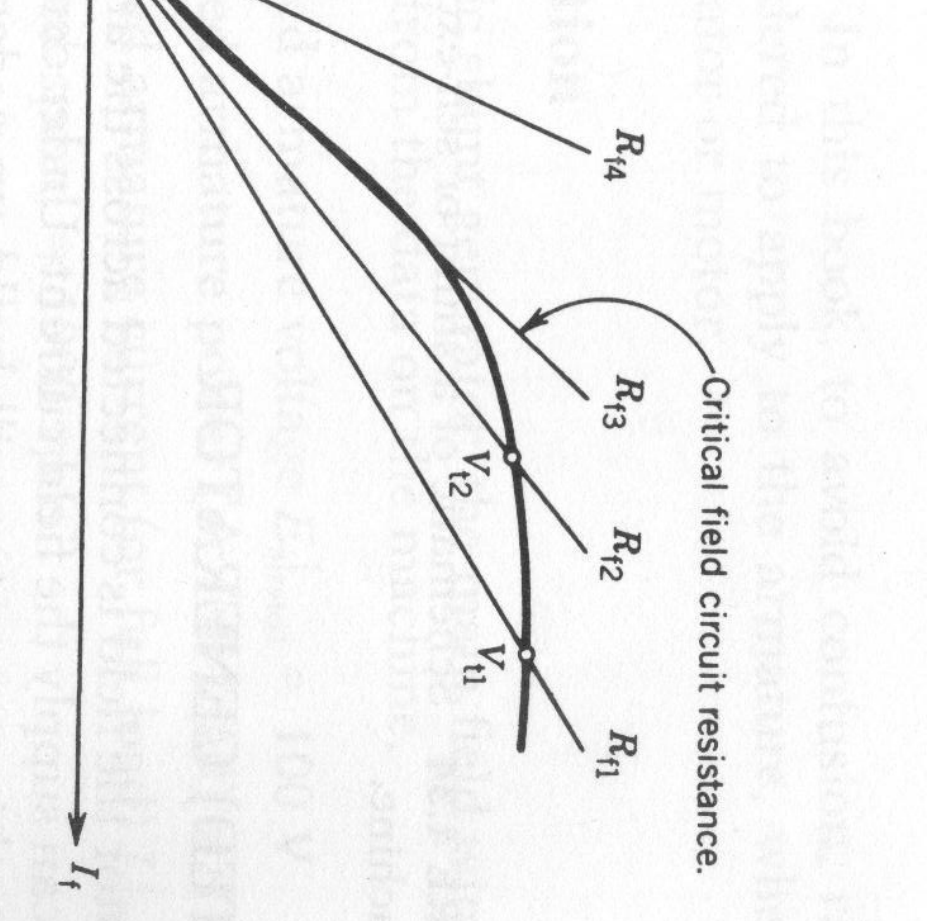
SCET Kodakara



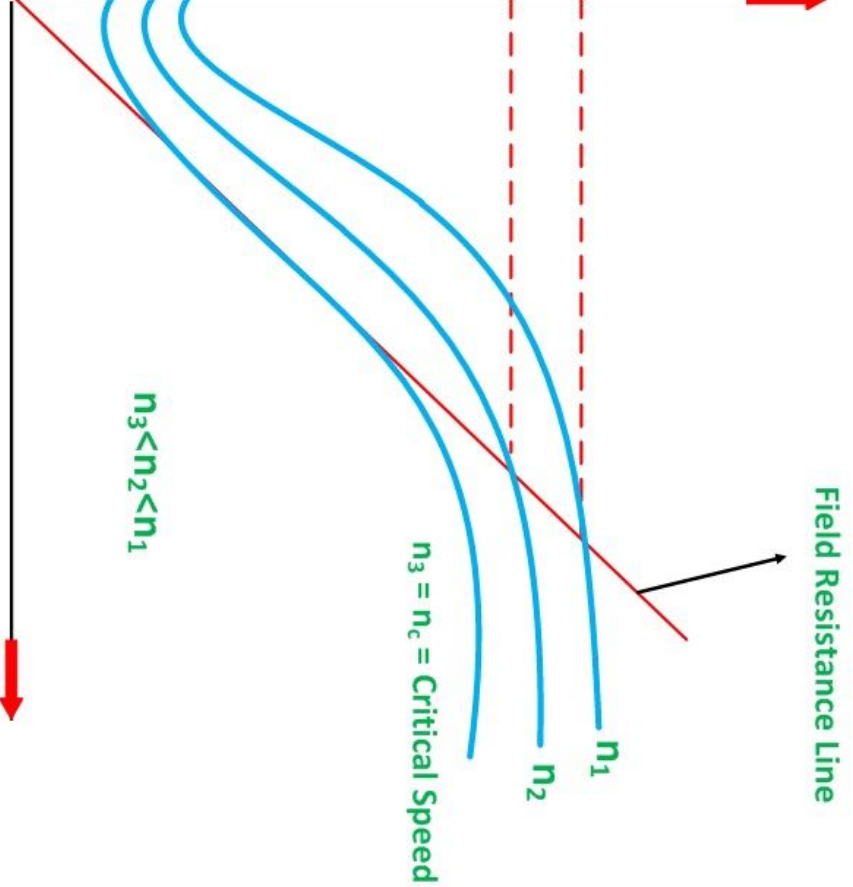
build-up in Shunt Generator



resistance (R_c): It is the maximum field resistance above which there will be no voltage in generator.



and (N_c): It is the minimum speed of
below which there will be no voltage



generator may fail to build up due to
ns:

magnetism: The voltage build up procedure can
e is residual magnetism. In case of new machine
has lost its magnetism due to aging, self excited
fail to build up the voltage. This difficulty can be
exciting the field winding from a separate DC
seconds with the armature at rest. This process
ashing.

ns reversed: The small voltage due to residual
uld circulate current in the field coils in such a
produce flux lines aiding the residual flux. If the
ns are incorrect, the flux produced by the small
pposes the residual flux and the generated
es to zero. This trouble can be remedied by
field connections with respect to armature

Circuit resistance: In case the field resistance is too high, due to a critical field resistance, voltage will not build up.

Below critical speed: Generator may fail to build up voltage due to the armature speed being less than the critical speed. This trouble can be remedied by increasing the speed above critical speed.

Problems

data pertain to the magnetization curve of a DC
at 1500rpm

1	0.8	1.2	1.6	2.0	2.4	2.8	3.0
120	172.5	202.5	221	231	237	240	

or, obtain

on open circuit to which the machine will
total shunt field resistance of 100Ω

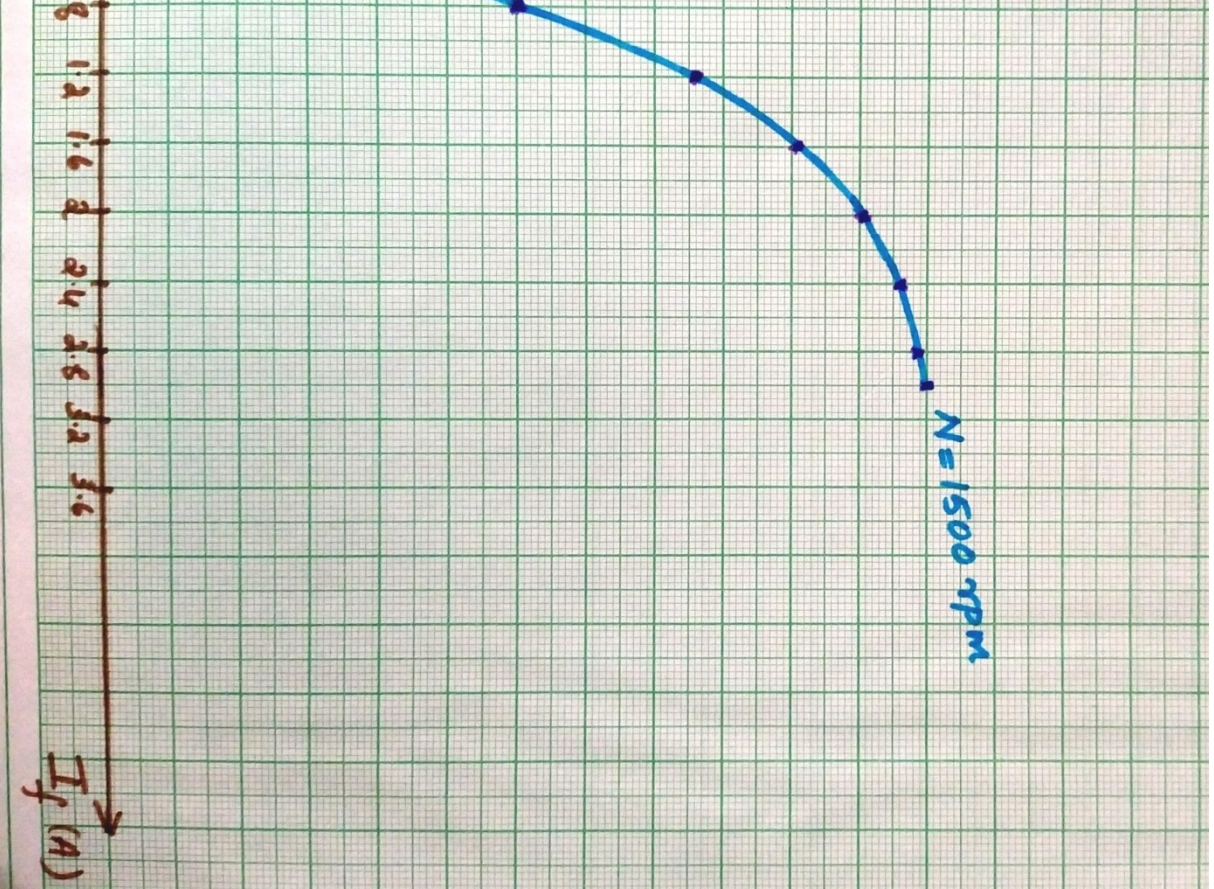
value of shunt field resistance at 1500rpm

speed for the shunt field resistance of 100Ω

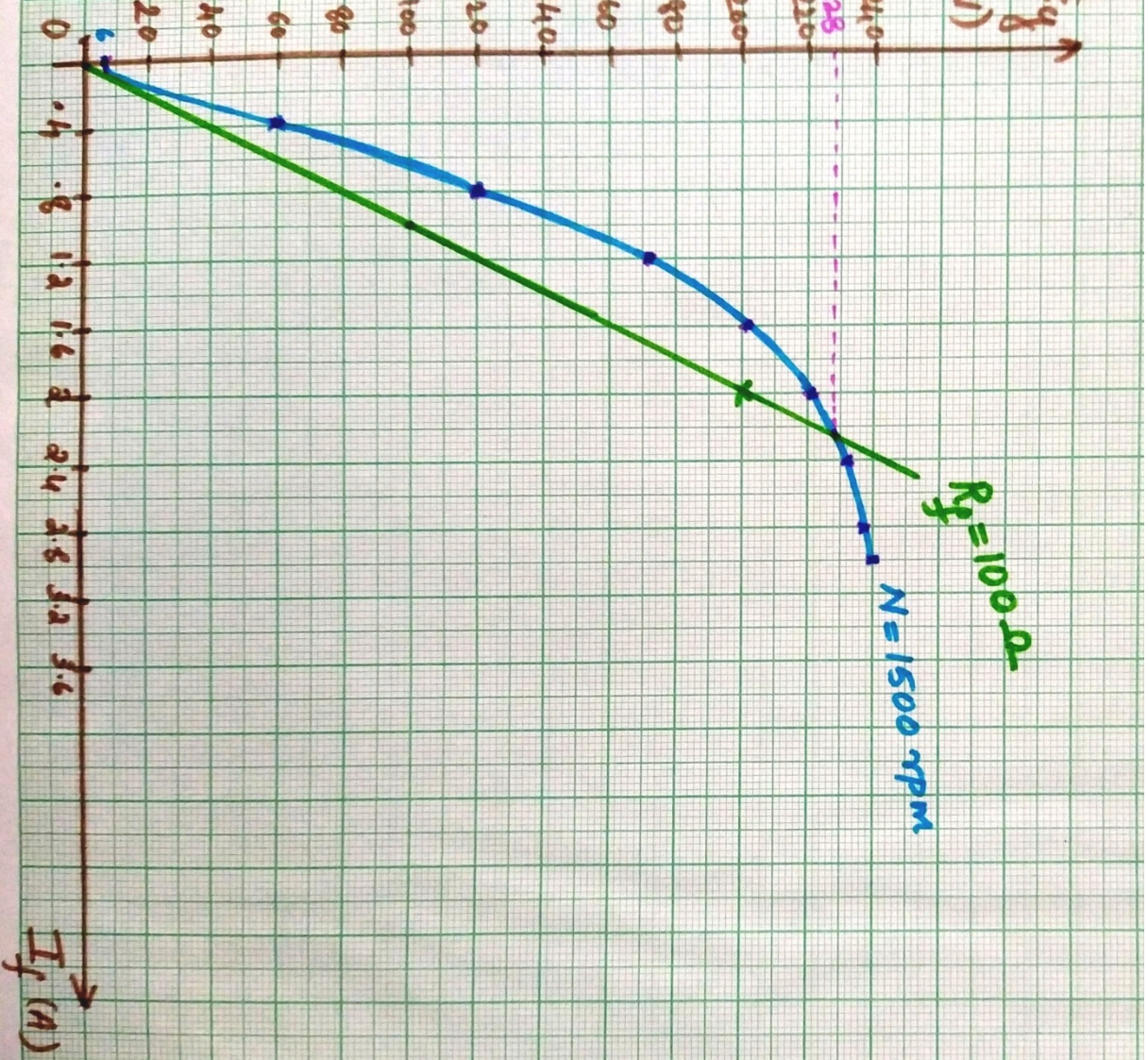
magnetization curve at 1200rpm and therefore the open

circuit for a field resistance of 100Ω

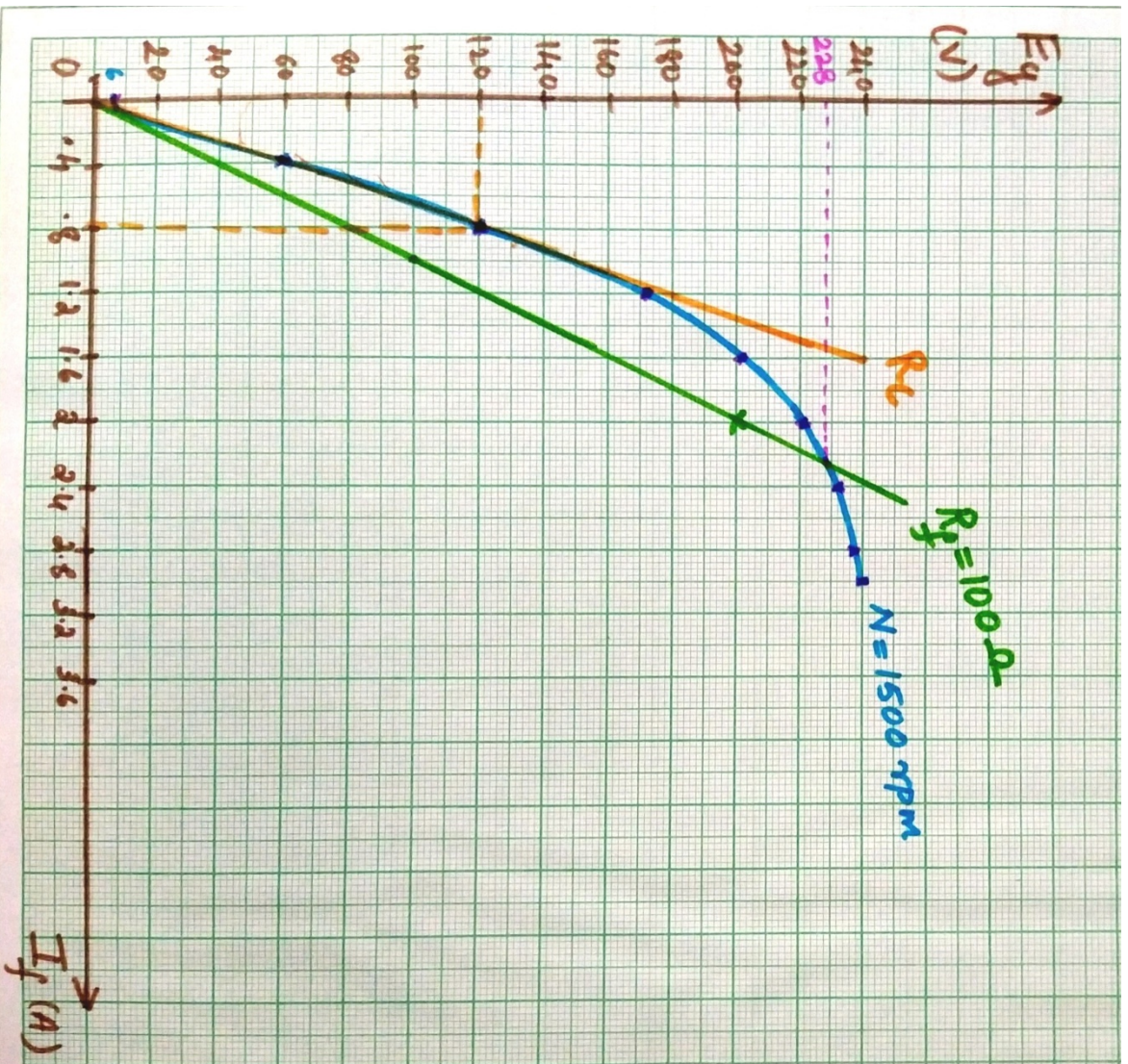
Problems



Problems

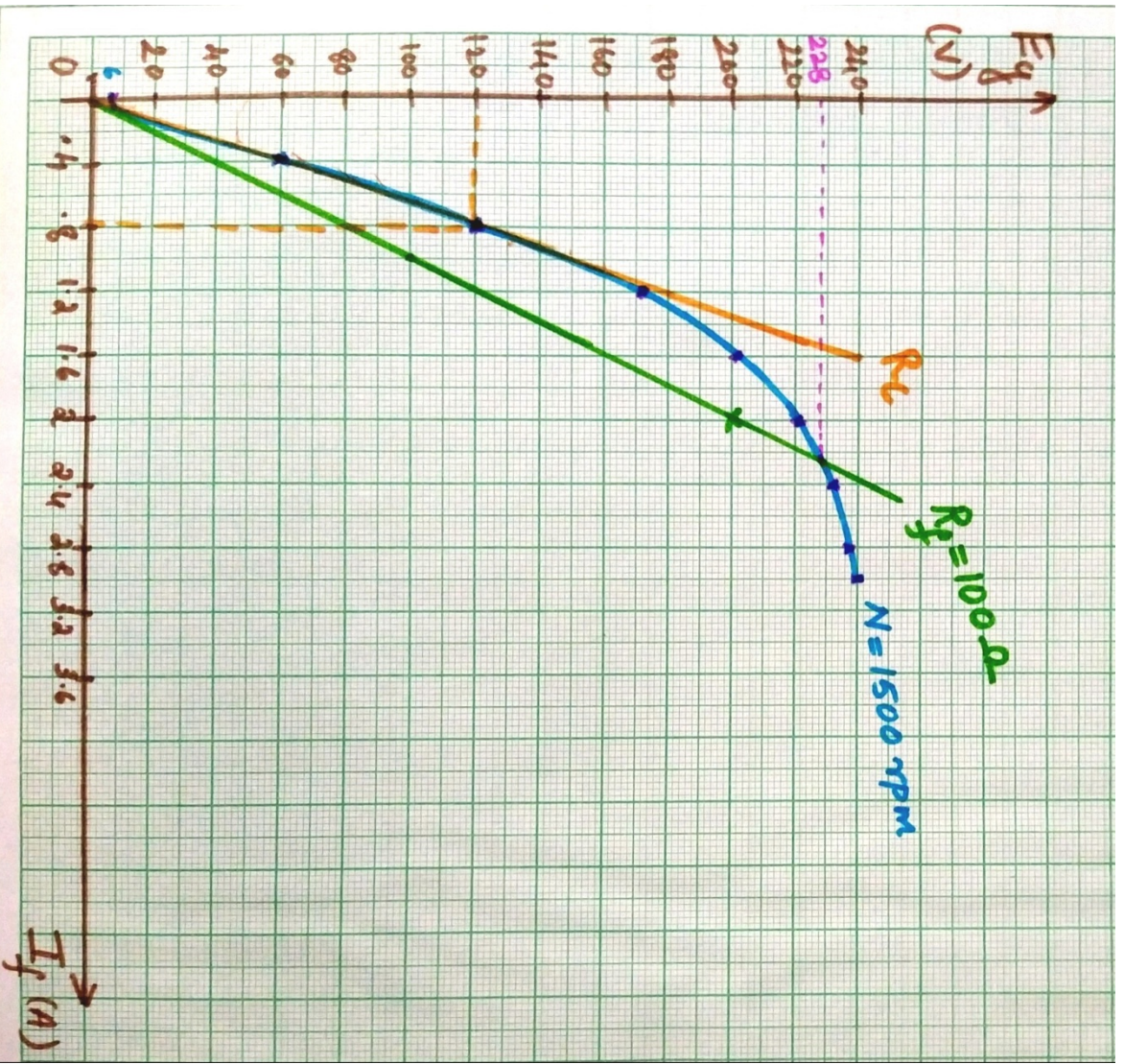


Problems

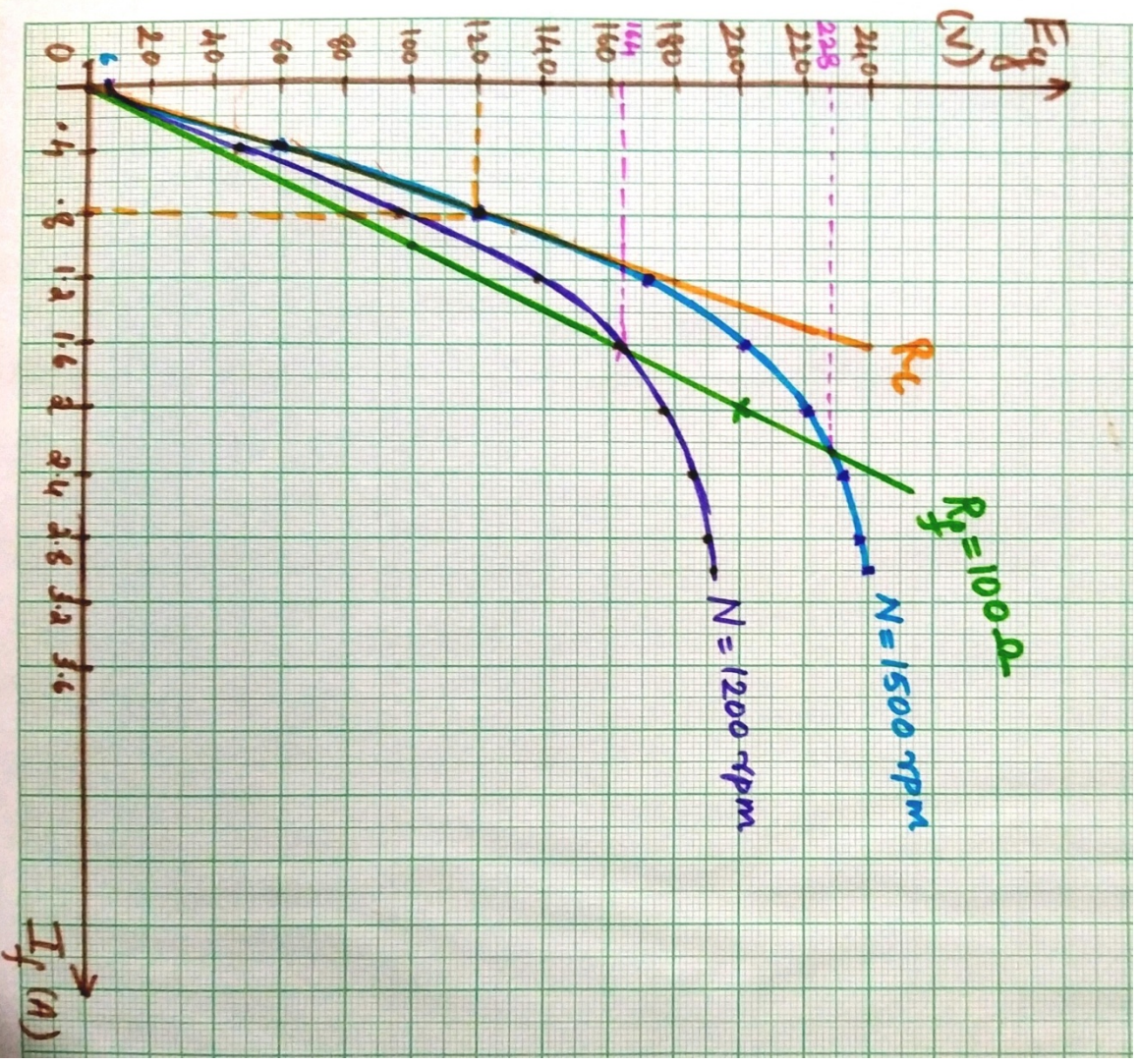


Problems

$= 1500$
 $= ?$



Problems



Problems

data pertain to the magnetization curve of a DC
at 1000rpm

0.2	0.4	0.6	0.8	1.0	1.2	1.4
50	100	140	170	190	200	205

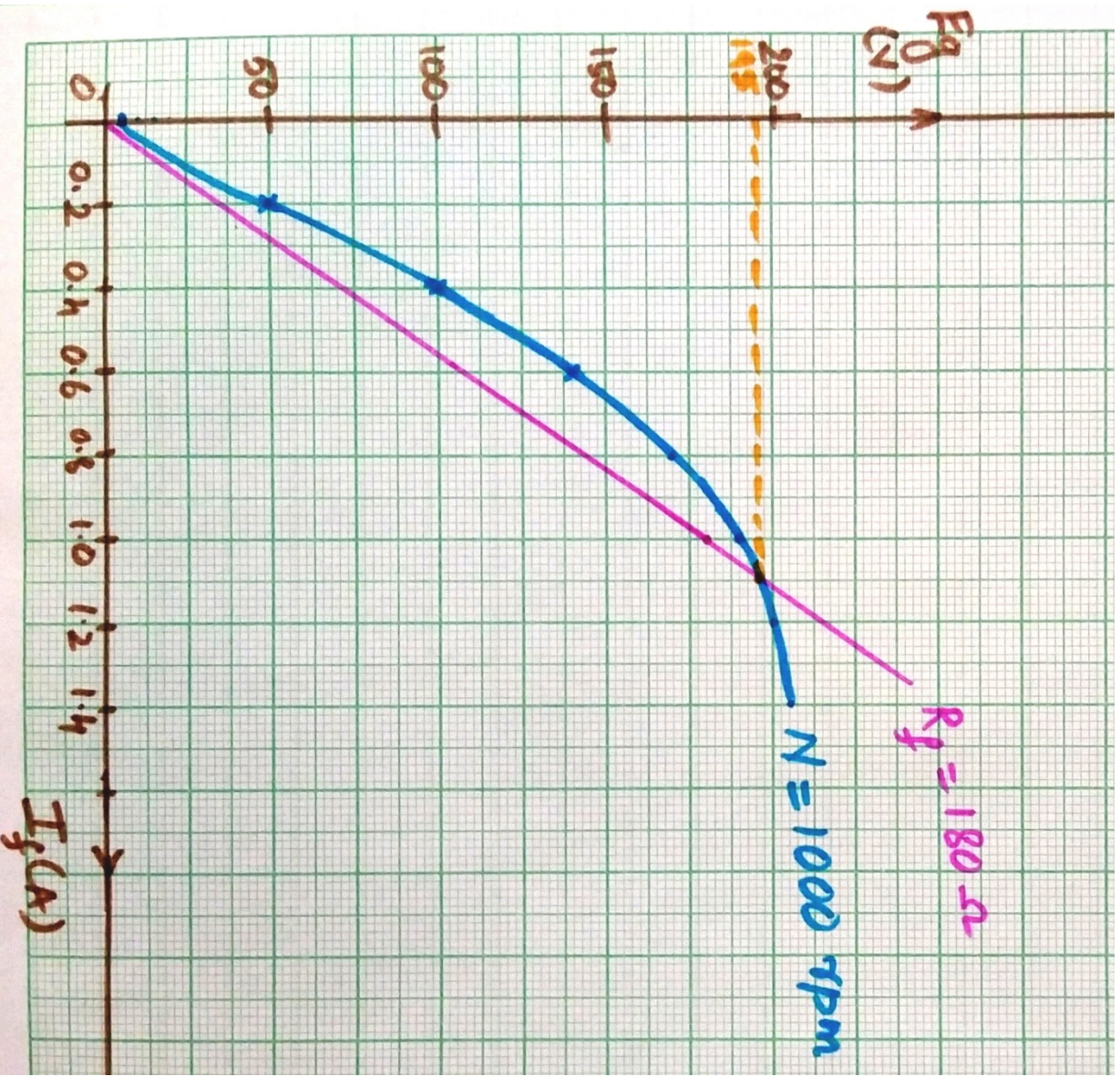
or, obtain

on open circuit to which the machine will
total shunt field resistance of 180Ω

value of shunt field resistance at 500rpm

speed for the shunt field resistance of 180Ω

Problems

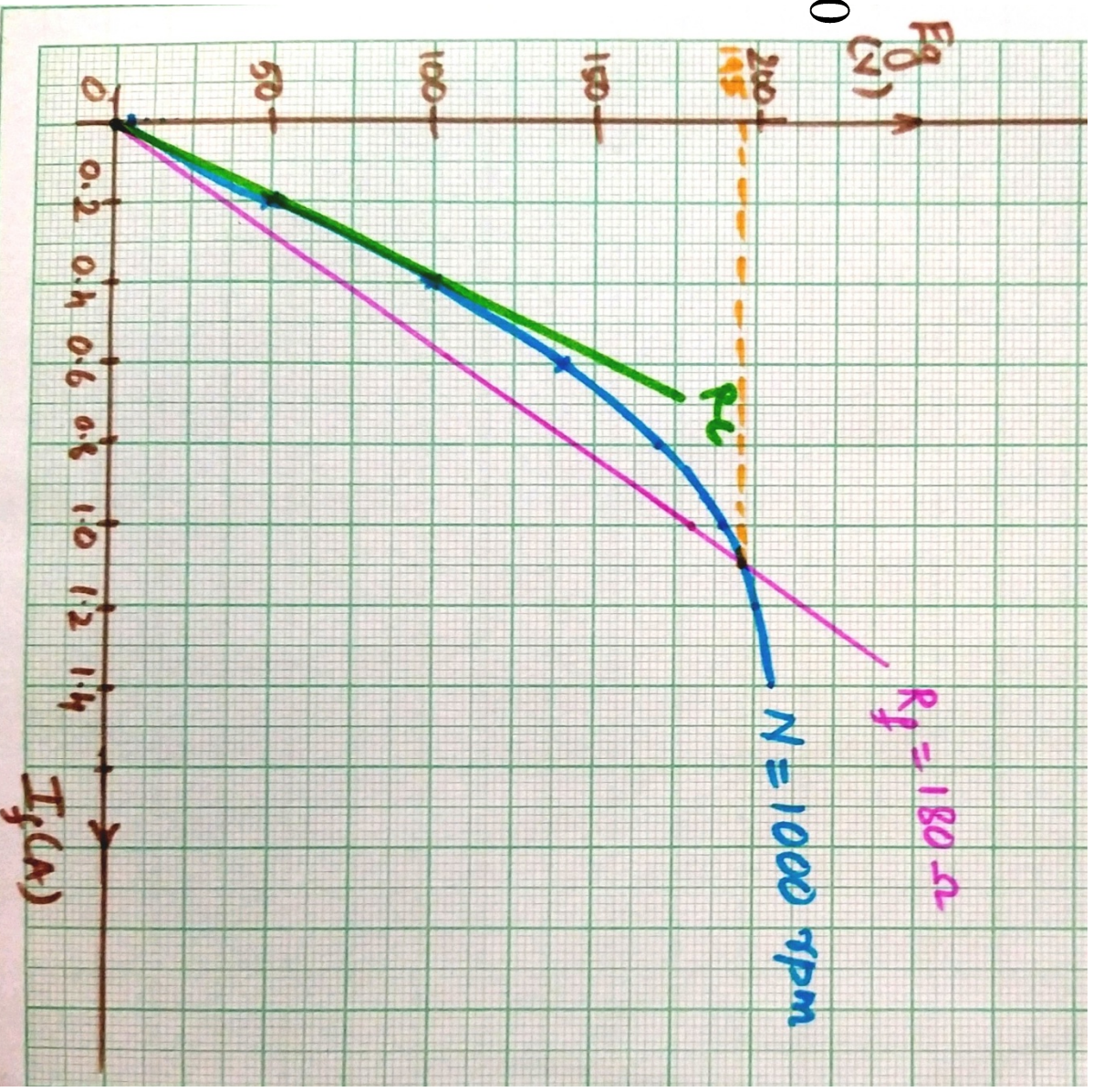


Problems

5

= 1000

00



Problems

$= 1000$
 $= ?$

